Surplus → Served — Project Scope & Requirement Document (Final, v1.0)

1. Project Overview

Name: Surplus → Served
Type: Mobile application (Android & iOS)
Category: Food donation / redistribution platform
Problem it solves: Cooked and surplus food from function halls, hostels, caterers, restaurants, and households goes to waste daily because there is no fast, structured way to connect that surplus with organizations that can use it before it spoils.
Solution: A three-sided platform — Donor, Receiver (verified NGOs), Admin — where a donor posts available food, the system automatically matches it to the best-positioned verified NGO nearby, and the NGO self-collects it within a fixed pickup window.
Core constraint driving every design decision: food has a short safe window (hours, not days). Every flow is built to minimize the time between "food is ready" and "food is collected."

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2. Objectives

1. Let a donor post surplus food in under a minute.
2. Get that post in front of the right nearby NGO within seconds, without manual admin work per post.
3. Guarantee a provable handover (no ambiguity about whether food was actually collected).
4. Support donors who have surplus every day (recurring), not just one-off events.
5. Serve Telugu- and English-speaking users equally.
6. Stay simple: no logistics, no liability system, no payments, no delivery layer in v1.

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3. Roles & Access

Role	Who	Registration gate
Donor	Function halls, hostels, caterers, restaurants, households	Phone OTP only — no document required
Receiver	Verified NGOs / registered organizations only	Phone OTP + one proof document + admin approval
Admin	Platform team	Internal accounts, no self-signup

No individual receivers, no volunteer/delivery role, no third role beyond these three — confirmed.

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4. Functional Requirements — Donor

4.1 Onboarding

- Sign up with name, phone (OTP verified), address
- No document upload required to start posting
- Language selection (English/Telugu) at first launch

4.2 Posting a donation

Required fields:

- Food name / description
- Category (cooked meal, groceries, bakery, etc.)
- Veg / non-veg
- Quantity in **servings** (not kg — servings is what an NGO plans around)
- Cooked-at date & time
- Best-before date & time
- Pickup window (start–end time)
- Pickup address
- Photo(s) — at least one required
- **Splittable?** toggle, with a minimum split unit if yes (see §7)
- **Recurring?** toggle, with frequency if yes (see §8)

4.3 Managing a live donation

- View real-time status: Live / Allocated / Collected / Reallocated / Unfulfilled / Cancelled
- See which NGO is currently allocated (name, contact) once matched
- Cancel a donation before it's allocated, with a one-tap reason
- Generate and display the **pickup code** once allocated
- Confirm collection is complete when the NGO enters the code

4.4 History & account

- List of past donations with status
- Basic count: total donations posted, total collected
- Edit profile / address / language
- Log out / delete account

4.5 Explicitly not in Donor scope

- CSR reports, tax receipts, impact certificates
- Choosing which NGO receives the donation (that's rule-based, §6)

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5. Functional Requirements — Receiver (NGO)

5.1 Onboarding & verification

- Org name, registration number (if available), contact person name, phone (OTP verified)
- Upload one proof document (registration certificate, or a signboard/ID photo as a lighter fallback for informal groups)
- Account stays in **Pending** state until admin approves
- Rejected applications show a reason and can be resubmitted
- No recurring re-verification — one-time approval; admin can revoke manually

5.2 Preferences setup

- Service radius (km)
- Food type preference (veg only / veg + non-veg)
- Typical intake capacity (servings)
- Preferred pickup time windows (optional, for filtering notifications)

5.3 Discovery & requesting

- Home feed of live donations within radius, nearest first
- Filter by food type, distance, quantity
- Push notification when a new donation matches preferences
- Tap **Request** to join the queue for a donation (does not guarantee allocation — see §6)

5.4 Once allocated

- Get exact pickup address, contact number, pickup window, and pickup code prompt
- In-app directions link (opens external maps app — no in-app navigation needed since there's no delivery tracking)
- Enter/show the donor's pickup code at the location to confirm collection
- Bring own containers (stated on every donation card as a reminder)

5.5 History & account

- List of past collections
- Rate the donor after each collection
- "Favourite" a donor to get priority notification (not priority allocation) on their future posts — useful for recurring donors
- Edit org profile, radius, preferences, language

5.6 Explicitly not in Receiver scope

- Individual (non-NGO) sign-up
- In-app chat/calling beyond a tap-to-call button
- Delivery/volunteer requests

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6. Functional Requirements — Admin

6.1 Verification queue

- List of pending NGO registrations with document preview
- Approve / Reject with reason
- Revoke an already-approved NGO if needed

6.2 Live operations

- Feed of all currently Live and Allocated donations
- Manual override: force-allocate, cancel, or reassign any donation
- View no-show / reallocation history per NGO

6.3 Rule engine configuration

Admin can tune, without needing a code deploy:

- Ranking formula weights (proximity, capacity match, request order, fairness)
- Pickup window default duration
- Notification radius rings and expansion timing
- Splittable donation threshold (minimum servings to qualify)

6.4 Disputes & basic moderation

- Flag/report a donor or NGO
- Suspend an account
- View a simple activity log per account (posts, collections, no-shows)

6.5 Dashboard

- Donations posted / live / collected / unfulfilled (counts, simple trend)
- Active donors, active NGOs
- No CSR/impact/export reporting in v1 (monetization-linked features are out of scope per your direction)

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7. Donation Lifecycle (state machine)

DRAFT
↓ donor submits
PENDING CHECK → REJECTED (missing photo / invalid dates)
↓ passes lightweight check
LIVE (visible to NGOs in radius, requests queue up)
↓ rule engine allocates ↓ donor cancels
ALLOCATED CANCELLED
↓ NGO enters pickup code ↓ pickup window expires
COLLECTED REALLOCATED → next NGO in queue
↓ queue exhausted
UNFULFILLED (donor notified, no penalty)

No in-transit/tracking state — since pickup is self-collect only, there is no delivery leg to track. Allocated → Collected is a direct jump, confirmed only by the pickup code.

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8. Allocation Logic — rules decide, no human in the loop by default

Triggered automatically the instant a donation goes Live, and again whenever a slot reopens (cancellation, no-show).

Ranking formula (admin-tunable weights, defaults below):

score = (proximity × 0.35) + (capacity match × 0.30)
+ (request order × 0.20) + (fairness × 0.15)

- **Proximity** — distance from NGO to donor address
- **Capacity match** — how well the NGO's stated intake fits the donation size
- **Request order** — first-come tiebreaker among near-equal scores
- **Fairness** — down-weights an NGO that won the last 1–2 allocations nearby, so no single NGO absorbs everything

No trust/safety scoring is built in — this stays out per your call on liability. Ratings are visible on profiles but do not block or boost allocation in v1.

Timers:

- Pickup window default: 45–60 minutes from allocation (admin-configurable)
- Notification radius: starts at 5 km, expands to 10 km if unclaimed after ~10 minutes (admin-configurable)
- On timeout: automatic reallocation to the next-ranked NGO in the queue, no admin action needed

Admin override exists for edge cases but is never the default path.

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9. Splitting a Donation

- Donor marks a post **splittable** at creation, with a minimum split unit (e.g., "no split below 5 servings")
- Only donations above the admin-configured threshold (suggested default: 20 servings) are eligible
- The rule engine can allocate different portions to different NGOs within the same Live cycle
- Each portion gets its own pickup code and its own pickup window
- The parent donation only reaches Collected once every portion is Collected or Unfulfilled
- If one portion times out, only that portion re-enters the queue — the rest are unaffected

- Non-splittable donations behave as a single all-or-nothing allocation

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10. Recurring Donations

- Donor sets a template once: food type, quantity range, pickup window, frequency (daily / specific weekdays / custom)
- Before each cycle, donor gets a **confirm-or-skip** notification — nothing auto-posts without a same-day confirmation, so a hostel mess isn't forced to donate on a day it has no surplus
- Each cycle still passes through the full Draft → Pending → Live → Allocated → Collected flow — recurring donors get no shortcut in the queue
- NGOs can favourite a recurring donor for priority notification (not priority allocation) on future posts

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11. Address Privacy

- While Live: donation shows only an approximate locality/area on the map
- On Allocation: exact address, contact person, and phone number unlock — visible only to the allocated NGO (or the specific NGO holding that portion, in a split donation)

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12. Handover Proof

- A 4-digit pickup code is generated the moment a donation (or a split portion) is Allocated
- Shown to the donor; the NGO representative enters or shows it on arrival to confirm collection
- This single code exchange is the entire proof mechanism for v1 — no photo requirement, no signature, no live GPS tracking (kept deliberately light since there's no liability layer)

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13. Containers

- Receiver (NGO) is responsible for bringing their own containers/vessels
- Stated as a persistent reminder line on every donation detail screen and in the pickup notification — no container logistics feature is built

14. Language

- Full parity in **English and Telugu** across every screen, push notification, and SMS
- Language toggle available at onboarding and in settings, switchable anytime
- No other languages in v1

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15. Non-Functional Requirements

- **Platform:** Native or cross-platform mobile app (Android + iOS), no web app in v1
- **Notifications:** Push (primary) — SMS fallback should be considered for NGOs on basic connectivity, though not mandatory for v1 launch
- **Performance:** Allocation must resolve within seconds of a donation going Live — this is a hard product requirement, not just a nice-to-have, given the food-safety clock
- **Availability:** No formal uptime SLA required for v1, but allocation and notification paths are the two flows that cannot be allowed to silently fail
- **Data retention:** Not specified yet — flagged as an open item (§18)

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16. Data Model (high level — for the dev team, not exhaustive)

- **User** (donor or NGO) → phone, name, language, address, role
- **NGORprofile** → org name, reg. number, proof document, verification status, radius, capacity, food preference
- **Donation** → donor ref, food details, quantity, splittable flag, recurring template ref, status, timestamps
- **DonationPortion** (only when split) → parent donation ref, quantity, status, allocated NGO ref, pickup code
- **RequestQueue** → donation/portion ref, NGO ref, requested-at timestamp, rank score
- **RecurringTemplate** → donor ref, food details template, frequency, next-confirm-due date
- **Rating** → from ref, to ref, donation ref, score, optional comment
- **AdminConfig** → rule weights, timer durations, radius rings, split threshold (single editable record)

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17. Explicitly Out of Scope for v1

- Volunteer / delivery portal or any transport layer
- Food safety, liability disclaimers, FSSAI-style compliance
- Individual (non-NGO) receivers
- Monetization, CSR reporting, sponsored donation drives, tax receipts
- Live GPS/delivery tracking (no delivery leg exists to track)
- In-app chat (tap-to-call only)
- Web application
- Languages beyond English and Telugu
- Trust-score-driven allocation (ratings are visible, not functional yet)

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18. Open Items Still Needing Your Decision

1. **SMS fallback for NGOs** — is push-only acceptable for v1, or do some target NGOs need SMS because of weak app usage/connectivity?
2. **Data retention** — how long should completed/expired donation records be kept?
3. **What happens to a donor's account if it repeatedly posts and gets no takers** — any flag, or purely informational?
4. **Rating visibility** — should an NGO's/donor's rating be visible to the other party before a request is made, or only after?